



INTERNAL OPERATIONS DOCUMENT — CONFIDENTIAL

Treemarkables Job Management Platform

SaaS Subscriber Onboarding Guide

A complete reference for setting up every third-party integration when onboarding a new subscriber to the job management platform. Includes account requirements, credentials, step-by-step setup, environment variables, and cost estimates.

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For internal use by the platform owner and designated operations staff only.

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1. Platform Overview

The **Treemarkables Job Management Platform** is a comprehensive, mobile-first business management system built specifically for New Zealand-based arborist and tree service companies. It is built on a modern web stack (React + Node.js + PostgreSQL) and includes a native iOS app via Apple Capacitor.

What the platform does

Module	Capability
Job Management	Full job lifecycle — leads, quotes, proposals, scheduling, dispatch, completion, invoicing. Dispatch board with ServiceM8-style layout. AI-powered smart scheduling.
Customer & Sales	Lead tracking, customer CRM, VIP membership, quote builder, interactive proposals with customer option selection, conversion analytics.
Scheduling	AI Smart Dispatch (GPT-4o), staff licence matching, equipment licence checks, daily revenue targeting, Google Calendar sync.
Communications	SMS (SMS Everyone NZ), transactional email (Resend), Gmail reply capture, Twilio voice calls + call recording, AI-generated quote messages.
Crew & Equipment	Staff management, licence/ticket tracking, equipment register, pre-start vehicle inspections, time tracking with charge-out rates.
Invoicing & Accounting	Invoice builder, line items from time tracking, Xero sync, GST-inclusive pricing in NZD.
Marketing	Mailchimp customer sync, Meta (Facebook/Instagram) ad campaign manager, automated review posting, marketing planner.
AI Features	AI chat assistant (GPT-4o with live business data), speech-to-quote (Whisper), AI dispatch, AI text extraction from Facebook messages.
Notifications	Push notifications via Firebase (job assignments, schedule changes, new leads, invoice payments), SMS alerts, email alerts.
iOS App	Native iOS app via Capacitor — CallKit integration with Twilio, push notifications via APNs/FCM, camera access, PWA install.

Module	Capability
Operations	Route optimisation, weather integration, mulch drops tracker, safety reporting, job diary, photo documentation.

Target subscriber

NZ-based arborist or tree service businesses with 2-20 staff. Ideally owner-operated businesses wanting to replace paper-based, spreadsheet, or ServiceM8 workflows with a purpose-built system.

2. Subscription & Infrastructure Model

How the platform is hosted

The platform currently runs as a single-tenant deployment per subscriber — each subscriber gets their own isolated environment with their own database and API credentials. This means there is no data co-mingling between subscribers, and each subscriber's credentials are fully separate.

i Current Architecture: Single codebase deployed per subscriber using environment variables to isolate credentials. Future versions may move to a multi-tenant architecture with shared hosting and per-subscriber database schemas.

What you manage centrally (Platform Owner)

- The codebase and deployments (Replit or other hosting)
- The iOS app on Apple App Store / TestFlight (one app, subscriber logs in)
- OpenAI API key (shared or per-subscriber with usage limits)
- Firebase project (one project, one service account, all subscriber devices register under it)
- Addy.co.nz API key (can be shared — rate limits per key)
- Resend sending domain (centralised domain, subscriber gets their own from-address)
- Platform support, updates, and bug fixes

What is set up per subscriber

- PostgreSQL database (separate Neon project)
- Xero Custom Connection (subscriber's own Xero organisation)
- SMS Everyone account (subscriber's business name as Sender ID)
- Twilio subaccount with a dedicated NZ phone number
- Gmail account for reply capture (or subscriber's own Gmail)
- Mailchimp audience (if subscriber uses email marketing)
- Meta page access token (if subscriber uses Facebook/Instagram marketing)
- Google Calendar OAuth (subscriber's own Google account)

Billing model suggestion

Fee type	Recommended approach
Platform subscription	Fixed monthly fee in NZD (see tiers section). You keep the margin between your fee and the actual third-party costs.
SMS costs	Either include a credit bundle in the plan, or pass through actual SMS Everyone costs with a small markup. SMS credits are pre-purchased.
AI usage	Include an allowance (e.g. 10,000 tokens/day) in the plan. Heavy AI users may need a higher tier.
Xero subscription	Subscriber pays Xero directly — this is their own accounting software. Not part of your platform fee.
Apple Developer fee	You pay the \$149 USD/year fee (shared across all subscribers). Amortise into your pricing.

3. Suggested Subscription Tiers

Starter	Professional 	Enterprise
\$149/mo NZD ✓ Job management & dispatch ✓ Customer & quote management ✓ SMS & email (100 SMS/mo included) ✓ Invoicing (no Xero sync) ✓ 2 staff accounts ✓ iOS app access ✓ Push notifications	\$299/mo NZD ✓ Everything in Starter ✓ Xero accounting sync ✓ Twilio voice + call recording ✓ AI Smart Dispatch ✓ AI chat assistant ✓ Google Calendar sync ✓ 500 SMS/mo included ✓ Up to 10 staff accounts ✓ Mailchimp sync	\$499/mo NZD ✓ Everything in Professional ✓ Unlimited staff accounts ✓ Meta ad campaign manager ✓ Unlimited SMS (fair use) ✓ Priority support ✓ Custom branding ✓ Dedicated onboarding call

 These suggested prices are indicative only. Review actual costs per subscriber (Section 7) before finalising pricing. Ensure your plan price covers third-party costs plus your labour and hosting overhead.

4. Integration Inventory

Every integration used by the platform is documented below. For each one you will find: what it does, who sets it up, what credentials are needed, and estimated monthly cost per subscriber.

PostgreSQL Database (Neon)

Required

Purpose: Stores all subscriber data — jobs, customers, staff, invoices, diary entries, settings, etc. Each subscriber must have their own isolated database.

Item	Detail
Account type	Neon serverless PostgreSQL (neon.tech) — one project per subscriber
Who creates	Platform owner (you)
Credentials needed	<code>DATABASE_URL</code> — full connection string from Neon dashboard
Free tier	Yes — 0.5 GB storage, 190 compute hours/month. Sufficient for small subscribers.
Paid tier	~NZD \$22/month for the Launch plan (10 GB storage, autoscaling)
Setup time	5 minutes — create project in Neon, copy connection string, run migrations

Setup steps

1. Go to **neon.tech** → Create account → New Project → select region closest to NZ (e.g. AWS ap-southeast-2 Sydney)
2. Copy the connection string from the project dashboard
3. Set it as `DATABASE_URL` in the subscriber's environment
4. Run `npm run db:push` to create all tables

Purpose: Syncs invoices, creates/matches customer contacts in Xero, pushes job invoices as accounts receivable (ACCREC) invoices. Uses Xero Custom Connection (machine-to-machine, no user OAuth required).

Item	Detail
Account type	Subscriber must have an active Xero organisation subscription (they pay Xero separately)
Who creates	You set up the Custom Connection app in the subscriber's Xero Developer Portal
Credentials needed	<code>XERO_CLIENT_ID</code> , <code>XERO_CLIENT_SECRET</code>
Token type	Client Credentials (expires every 30 minutes — auto-refreshed by the platform)
Cost to you	\$0 — Xero API access is free. Subscriber pays Xero for their plan (~\$50-100/month).
Setup time	20 minutes

Setup steps

1. Log in to **developer.xero.com** using the subscriber's Xero account
2. Go to **My Apps → New App**
3. Select **Custom Connection** (not standard/partner)
4. Grant scopes: `accounting.contacts` `accounting.transactions` `accounting.settings`
(add payroll scopes if needed)
5. Copy Client ID and Client Secret into the subscriber's environment
6. In the platform Settings → Xero, click "Connect to Xero" to verify the connection
7. Set the Xero Sales Account Code (default: **200**) and Tax Type (default: **OUTPUT2** for 15% GST)

⚠ Custom Connection tokens expire every 30 minutes. The platform auto-refreshes them. However, if the `XERO_CLIENT_ID` / `XERO_CLIENT_SECRET` are rotated or the scopes change in Xero Developer Portal, you must hit the "Refresh Token" button in Settings → Xero to get a new token with updated scopes.

Purpose: Sends SMS messages to customers — booking confirmations, quote follow-ups, appointment reminders, marketing campaigns. Also captures customer SMS replies and logs them to the job diary.

Item	Detail
Account type	SMS Everyone NZ business account (smseveryone.co.nz)
Who creates	You create on behalf of subscriber, or subscriber creates themselves
Credentials needed	<code>SMSEVERYONE_USERNAME</code> , <code>SMSEVERYONE_PASSWORD</code> , <code>SMSEVERYONE_SENDER_ID</code>
Sender ID	Business name shown to recipients (e.g. "Treemarkables"). Must be approved — allow 1-3 business days.
Cost per SMS	~\$0.09-0.12 NZD per SMS (bulk discounts available)
Pre-purchase	Credits are pre-purchased. Recommend starting with 500-1000 credit bundle.
Setup time	30 minutes (plus 1-3 days for Sender ID approval)

Setup steps

1. Go to **smseveryone.co.nz** → Sign up for a business account
2. Verify email and complete business registration
3. Apply for a custom Sender ID (subscriber's business name) — takes 1-3 business days
4. Purchase SMS credits (start with at least 500)
5. Get API username and password from account settings
6. Set `SMSEVERYONE_USERNAME` , `SMSEVERYONE_PASSWORD` , `SMSEVERYONE_SENDER_ID` in subscriber environment

i The platform tries the `.co.nz` endpoint first, then falls back to `.com` . Both endpoints use the same credentials.

Purpose: Provides a dedicated NZ phone number for the subscriber. Customer calls are forwarded to Twilio, which records the call, transcribes it via Whisper AI, and optionally rings the iOS app via CallKit (Twilio Voice SDK). The subscriber's personal mobile rings via carrier forwarding — customers never know Twilio is involved.

Item	Detail
Account type	Twilio main account or subaccount (recommended: one subaccount per subscriber)
Who creates	Platform owner (you) — create a Twilio subaccount under your main account
Credentials needed	<code>TWILIO_ACCOUNT_SID</code> , <code>TWILIO_AUTH_TOKEN</code> , <code>TWILIO_API_KEY</code> , <code>TWILIO_API_KEY_SECRET</code> , <code>TWILIO_PHONE_NUMBER</code> , <code>TWILIO_TWIML_APP_SID</code>
NZ phone number cost	~\$2.50 USD/month for a local NZ number
Voice call cost	~\$0.013 USD/minute to receive calls
Call recording	Included — stored in Twilio for 90 days by default
Setup time	45-60 minutes

Setup steps

1. Log in to **twilio.com/console** → Create a Subaccount for the subscriber
2. Note the subaccount SID and Auth Token
3. Buy a New Zealand phone number in the subaccount
4. Create an API Key & Secret (recommended over Auth Token for production)
5. Create a TwiML App with the platform's voice webhook URL: `https://[subscriber-domain]/api/twilio/voice`
6. Note the TwiML App SID
7. Set all credentials in the subscriber's environment
8. In Settings → Business, enter the subscriber's personal phone number for call forwarding
9. Subscriber sets up carrier call forwarding: personal mobile → Twilio NZ number

 **Important:** The Twilio webhook secret (`HERO_WEBHOOK_SECRET` in the current deployment) must be set to validate Twilio webhook signatures. Do not skip this step.

📧 Resend (Transactional Email)

Required

Purpose: Sends all outgoing emails — quote/proposal emails, invoice emails, booking confirmations, general customer communications. Emails are sent from a verified domain (e.g. `info@updates.treemarkables.co.nz`) with the subscriber's business name as the sender.

Item	Detail
Account type	Resend.com account — free up to 3,000 emails/month and 100/day
Who creates	Platform owner (you) — can use a single Resend account with multiple sending domains
Credentials needed	<code>RESEND_API_KEY</code>
Domain verification	Add DNS records to the subscriber's sending domain (e.g. <code>updates.subscriberdomain.co.nz</code>)
Free tier	3,000 emails/month — sufficient for most small subscribers
Paid tier	~\$22 NZD/month for 50,000 emails/month (Pro plan)
Setup time	15 minutes (plus DNS propagation: up to 48 hours)

Setup steps

1. Log in to **resend.com** → Add a Sending Domain for the subscriber
2. Add the provided DNS records (TXT + MX) to the subscriber's domain registrar
3. Wait for DNS propagation (usually 15-60 minutes, up to 48 hours)
4. Verify the domain in Resend dashboard
5. Create an API key scoped to the verified domain
6. Update the `fromEmail` value in `server/resendClient.ts` with the subscriber's business name and domain
7. Set `RESEND_API_KEY` in subscriber environment

Purpose: The platform monitors a Gmail inbox via IMAP and automatically captures customer email replies. Captured replies are logged to the relevant job diary as diary entries, and linked to the customer record. Outgoing emails can also be sent via Gmail/SMTP as a fallback.

Item	Detail
Account type	Standard Gmail or Google Workspace account
Who creates	Subscriber (their own Gmail), or you create a dedicated platform Gmail for them
Credentials needed	<code>GMAIL_USER</code> (email address), <code>GMAIL_APP_PASSWORD</code> (16-character App Password)
2FA requirement	Google account must have 2-Factor Authentication enabled before App Password can be generated
Cost	Free (standard Gmail) or from \$8 USD/month for Google Workspace
Setup time	10 minutes

Setup steps

1. Log in to **myaccount.google.com** with the subscriber's Gmail account
2. Go to **Security → 2-Step Verification** → Enable it
3. Go to **Security → App passwords** → Create a new App Password
4. Select App: "Mail", Device: "Other (Custom name)" → enter "Treemarkables Platform"
5. Copy the 16-character App Password
6. Set `GMAIL_USER` = Gmail address, `GMAIL_APP_PASSWORD` = App Password in subscriber environment
7. Enable IMAP in Gmail Settings → See All Settings → Forwarding and POP/IMAP → Enable IMAP

Purpose: Powers the AI Business Assistant (GPT-4o with function calling for live business data), Speech-to-Quote (Whisper audio transcription), AI Smart Dispatch (GPT-4o schedule optimisation), and Facebook message text extraction.

Item	Detail
Account type	OpenAI API account (platform.openai.com)
Who creates	Platform owner (can share one key across subscribers, or create per-subscriber)
Credentials needed	OPENAI_API_KEY
Models used	GPT-4o (chat & dispatch), Whisper (audio transcription)
Cost estimate	~\$5-20 NZD/month per active subscriber depending on AI usage frequency
Usage limits	Set per-key spending limits in OpenAI dashboard to prevent runaway costs
Setup time	5 minutes

Setup steps

1. Log in to **platform.openai.com** → API Keys → Create new secret key
2. Set a monthly spend limit for the key (recommended: \$20-50 USD/subscriber)
3. Set `OPENAI_API_KEY` in subscriber environment
4. Optionally, use one shared key and monitor usage by Project in the OpenAI dashboard

i The platform uses **GPT-4o** (not temperature-controlled). Do not use older models — GPT-5+ does not accept the `temperature` parameter and will error if passed.

Purpose: Sends real-time push notifications to the iOS (and Android) app — job assignments, schedule changes, new leads, invoice payments, quote acceptances. Uses Firebase Admin SDK with a service account JSON.

Item	Detail
Account type	Firebase project (console.firebase.google.com) — can be shared across subscribers
Who creates	Platform owner — one Firebase project for all subscribers (or per-subscriber for isolation)
Credentials needed	<code>FIREBASE_SERVICE_ACCOUNT</code> (full JSON service account as a single-line string)
APNs key	Apple Push Notification service key must be uploaded to Firebase for iOS notifications
Cost	Firebase Spark (free) plan supports up to 1M notifications/month. Sufficient for all small subscribers.
Setup time	30 minutes

Setup steps

1. Go to **console.firebase.google.com** → Select or create a project
2. Go to **Project Settings** → **Service Accounts** → Generate new private key
3. Download the JSON file → minify it to a single line (remove newlines except in `private_key`)
4. Set as `FIREBASE_SERVICE_ACCOUNT` in subscriber environment
5. Go to **Project Settings** → **Cloud Messaging** → **Apple app configuration**
6. Upload the APNs authentication key (Key ID **D44XQ29XZM**, Team ID **7Q57T2KAH9** for current deployment)
7. Register the iOS app with bundle ID `com.treemarkables.app`

⚠ The APNs key is linked to the Apple Developer account. If you onboard a subscriber with a different bundle ID, you need a new Firebase app registration and potentially a new APNs key. APNs keys do not expire but can be revoked — keep them secure.

Purpose: Powers the address autocomplete field in job and customer forms. Searches NZ addresses in real-time using the Addy Solutions API. Falls back to mock data if the API is unavailable.

Item	Detail
Account type	Addy.co.nz API account
Who creates	Platform owner (can share one key across subscribers)
Credentials needed	<code>ADDY_API_KEY</code>
Cost	Pay-per-lookup — ~\$0.01-0.02 NZD per address search. Estimate \$3-10/month per active subscriber.
Rate limits	Shared key has shared rate limits. Separate keys per subscriber recommended for heavy volume.
Setup time	5 minutes

Setup steps

1. Go to **addy.co.nz** → Sign up for a developer account
2. Generate an API key in the account dashboard
3. Set `ADDY_API_KEY` in subscriber environment
4. The address search in the platform currently filters to Gisborne territory — update the territory parameter in `server/services/addressService.ts` to match the subscriber's region

⚠ The current codebase filters address searches to **Gisborne** territory only. For subscribers outside Gisborne, update the `territory` parameter in `server/services/addressService.ts` or remove it to search nationwide.

Purpose: Automatically syncs scheduled jobs to the subscriber's Google Calendar. Creates calendar events when jobs are scheduled, updates them when rescheduled, and removes them when cancelled. Staff assignments and job details are included in the event description.

Item	Detail
Account type	Standard Google account
Who creates	Subscriber authorises their own Google account via OAuth in the platform settings
Credentials needed	OAuth access token — managed via Replit Google Calendar Connector (no manual credential setup)
Cost	Free
Token expiry	Access tokens expire every hour — automatically refreshed by the platform
Setup time	5 minutes (subscriber authorises their own account)

Setup steps

1. In the platform's Replit environment, connect the Google Calendar integration via the Integrations panel
2. Subscriber logs into their Google account and grants calendar access
3. Calendar sync is immediately active — test by scheduling a job

i All calendar events are created in NZ timezone (Pacific/Auckland). The subscriber's calendar must also be set to NZ timezone to display times correctly.

Purpose: Syncs the subscriber's customer list to a Mailchimp audience. Customers are synced with name, email, phone, and address data. Allows the subscriber to run email marketing campaigns directly from Mailchimp using their customer data.

Item	Detail
Account type	Mailchimp account (subscriber's own)
Who creates	Subscriber creates their own Mailchimp account
Credentials needed	<code>MAILCHIMP_API_KEY</code> , <code>MAILCHIMP_AUDIENCE_ID</code> (set in platform Business Settings)
Cost	Free for up to 500 contacts. \$25–75 USD/month for larger lists.
GDPR/Privacy	Subscriber is responsible for customer consent for marketing emails
Setup time	15 minutes

Setup steps

1. Subscriber creates or logs into **mailchimp.com**
2. Go to **Account** → **Extras** → **API keys** → Create a key
3. Note the Audience ID from **Audience** → **Manage Audience** → **Settings** → **Audience name and defaults**
4. In the platform Settings → Marketing, enter the API Key and Audience ID
5. Click "Sync Customers" to do the initial sync

Purpose: Manages Facebook and Instagram ad campaigns directly from the platform. Can create campaigns, monitor performance, and automatically post customer reviews to the subscriber's Facebook page. Uses the Meta Graph API.

Item	Detail
Account type	Facebook Business Manager account with a Facebook Page and Ad Account
Who creates	Subscriber creates their own Facebook Business/Ad account
Credentials needed	<code>FACEBOOK_PAGE_ACCESS_TOKEN</code> , <code>FACEBOOK_PAGE_ID</code> , <code>FACEBOOK_AD_ACCOUNT_ID</code>
Token type	Page Access Token — long-lived (never expires if extended properly)
Ad spend	Subscriber's own ad budget — paid to Meta. Not included in your platform fee.
Setup time	30-60 minutes

Setup steps

- Subscriber goes to **developers.facebook.com** → Create App → Business type
- Add "Facebook Login" and "Pages API" products
- Generate a short-lived user token, then exchange for a Page Access Token using the Graph API Explorer
- Extend the token to a long-lived token (60 days, or use System User token for never-expiring)
- Find the Page ID from the Facebook Page's About section
- Find the Ad Account ID from Facebook Business Manager → Ad Accounts
- Set all three credentials in subscriber environment

⚠ Page Access Tokens expire after 60 days unless converted to System User tokens. Set a calendar reminder to refresh tokens, or use a System User token (never expires) created in Facebook Business Manager.

Purpose: Distributes the native iOS app (built with Capacitor) to subscribers via TestFlight (beta testing) or the App Store. The APNs authentication key from this account is also required for Firebase push notifications to work on iOS.

Item	Detail
Account type	Apple Developer Program — individual or organisation
Who creates	Platform owner (you) — one developer account serves all subscribers
Current bundle ID	<code>com.treemarkables.app</code>
Current Team ID	<code>7Q57T2KAH9</code>
APNs Key ID	<code>D44XQ29XZM</code>
Annual cost	\$149 USD/year (~\$250 NZD/year) — amortise across subscribers
TestFlight invite	Each subscriber is added as a TestFlight external tester via their Apple ID
Setup time	App Store review takes 1-3 days for new builds. TestFlight: hours.

Per-subscriber setup steps

1. Get subscriber's Apple ID email address
2. In **App Store Connect** → TestFlight → External Testers → Add subscriber's Apple ID
3. Subscriber accepts TestFlight invitation on their iPhone → installs the app
4. Subscriber logs in with their platform username/password (not Apple ID)
5. Push notification permissions are requested on first launch

i For full App Store distribution (not TestFlight), the app must pass Apple's review. The current app is Treemarkables-branded — for whitelabelled subscribers you may need a separate app binary and App Store listing.

Purpose: Stores job site photos uploaded via the iOS app or web dashboard. Photos are stored with automatic thumbnail generation. Served via public URLs for in-app display.

Item	Detail
Account type	Managed via Replit Object Storage (backed by Google Cloud Storage)
Who creates	Platform owner — enable via Replit Object Storage in the project settings
Credentials needed	Automatically managed by Replit — no manual credential setup required
Storage structure	<code>public/</code> for served assets, <code>.private/</code> for private files
Cost	Included in Replit plan for small usage. ~\$0.023 USD per GB stored per month for larger volumes.
Setup time	2 minutes — enable in Replit project settings

Setup steps

1. In the Replit project → click **Object Storage** in the left panel
2. Click **Create Bucket** — it creates automatically with environment variables set
3. No further configuration needed — photo upload APIs use the bucket automatically

5. New Subscriber Onboarding Checklist

Use this checklist for every new subscriber sign-up. Work through each step in order. Estimated total setup time: **3-5 hours** (spread over 1-3 business days due to DNS propagation and Sender ID approval).

Day 1 — Core Infrastructure

1 Provision database

5 min

Create a new Neon project → copy DATABASE_URL → run `npm run db:push` to create all tables

2 Deploy subscriber environment

15 min

Fork/copy the Replit project for the subscriber → set DATABASE_URL → test that the app starts

3 Configure Resend email domain

15 min + wait

Add subscriber's sending domain in Resend → get DNS records → add DNS records at their registrar → wait for propagation

4 Set up Gmail reply capture

10 min

Enable 2FA on Gmail → generate App Password → enable IMAP → set GMAIL_USER and GMAIL_APP_PASSWORD

5 Set OpenAI API key

5 min

Create or reuse API key with spending limit → set OPENAI_API_KEY

6 Set Firebase service account

15 min

Download service account JSON → minify → set FIREBASE_SERVICE_ACCOUNT → register subscriber's iOS devices in Firebase

7 Enable Object Storage

5 min

Enable Replit Object Storage for the subscriber's project → verify photos can upload

8 Register subscriber on TestFlight

10 min

Add subscriber's Apple ID as a TestFlight external tester → they accept invite and install the app on iPhone

9 Apply for SMS Sender ID

30 min + wait

Register SMS Everyone account → apply for custom Sender ID using subscriber's business name (1-3 days approval)

Day 2 — Communications & Accounting

- 10 Verify Resend domain** 10 min

Confirm DNS records have propagated → verify domain in Resend → generate API key → set RESEND_API_KEY
- 11 Configure Twilio voice (Pro+ only)** 45 min

Create Twilio subaccount → buy NZ number → create TwiML App → create API Key → configure all Twilio env vars → instruct subscriber to set up carrier call forwarding
- 12 Set up Xero connection (Pro+ only)** 20 min

Log into subscriber's Xero Developer Portal → create Custom Connection app → grant accounting scopes → copy Client ID & Secret → test connection in platform Settings → Xero
- 13 Set Addy.co.nz region** 5 min

Update territory parameter in addressService.ts to match subscriber's region → set ADDY_API_KEY
- 14 Connect Google Calendar (Pro+ only)** 5 min

Subscriber authorises their Google account via the platform's integrations panel → test by scheduling a job and checking calendar

Day 3 — Data Entry & Handover

- 15 Activate SMS credentials** 10 min

Confirm Sender ID approval → set SMSEVERYONE_USERNAME, SMSEVERYONE_PASSWORD, SMSEVERYONE_SENDER_ID → send a test SMS
- 16 Configure business settings** 30 min

In platform Settings: business name, address, GST number, daily revenue target, invoice payment terms, job categories, staff members with charge-out rates
- 17 Import existing customers & jobs (if any)** Variable

If migrating from ServiceM8 or spreadsheets, import customer and job data via API or manual entry
- 18 Configure Mailchimp (Pro+, optional)** 15 min

Subscriber provides Mailchimp API key and audience ID → enter in Settings → Marketing → do initial customer sync
- 19 Configure Meta (Enterprise, optional)** 45 min

Generate and extend Facebook Page Access Token → get Page ID and Ad Account ID → set in environment
- 20 Subscriber handover & training** 1-2 hours

Walkthrough call: add first job, schedule it, send a quote, create an invoice. Provide access credentials and support contact.

6. Environment Variables Master Reference

Every environment variable used by the platform, grouped by integration. Column "Set by" indicates whether you (the platform owner) set this variable, or whether it comes from the subscriber.

Database		
Variable	Description	Set by
DATABASE_URL	Full PostgreSQL connection string (Neon). Format: <code>postgresql://user:pass@host/dbname?sslmode=require</code>	Owner (per subscriber)

Xero Accounting		
Variable	Description	Set by
XERO_CLIENT_ID	Client ID from Xero Custom Connection app in subscriber's Xero Developer Portal	Owner (per subscriber)
XERO_CLIENT_SECRET	Client Secret from the same Xero Custom Connection app	Owner (per subscriber)

SMS — SMS Everyone NZ		
Variable	Description	Set by
SMSEVERYONE_USERNAME	SMS Everyone account username	Owner (per subscriber)
SMSEVERYONE_PASSWORD	SMS Everyone account password	Owner (per subscriber)
SMSEVERYONE_SENDER_ID	Approved sender name shown to recipients (e.g. "Treemarkables")	Owner (per subscriber)

Telephony — Twilio		
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Variable	Description	Set by
TWILIO_ACCOUNT_SID	Twilio subaccount SID (starts with AC)	Owner (per subscriber)
TWILIO_AUTH_TOKEN	Twilio subaccount auth token (use API Key instead for production)	Owner (per subscriber)
TWILIO_API_KEY	Twilio API Key SID (starts with SK) — preferred over Auth Token	Owner (per subscriber)
TWILIO_API_KEY_SECRET	Twilio API Key Secret (used in twilioClient.ts for authentication)	Owner (per subscriber)
TWILIO_PHONE_NUMBER	The Twilio NZ phone number (e.g. +6499999999)	Owner (per subscriber)
TWILIO_TWIML_APP_SID	Twiml App SID for voice SDK (starts with AP)	Owner (per subscriber)
HERO_WEBHOOK_SECRET	Secret for validating Twilio webhook signatures	Owner (per subscriber)
HERO_PHONE_NUMBER	Subscriber's personal mobile number for call forwarding	Subscriber
TELNYX_API_KEY	Telnyx fallback telephony (if used instead of Twilio)	Owner (optional)
VONAGE_NUMBER	Vonage backup number (if used)	Owner (optional)
VONAGE_FORWARD_TO_NUMBER	Number to forward Vonage calls to	Owner (optional)

Email — Resend

Variable	Description	Set by
RESEND_API_KEY	Resend API key (starts with re_)	Owner (shared or per subscriber)

Email — Gmail Reply Capture

Variable	Description	Set by
EMAIL_USER	Gmail address to monitor for customer replies	Owner or Subscriber
EMAIL_APP_PASSWORD	16-character Google App Password (not the account password)	Owner or Subscriber

AI — OpenAI

Variable	Description	Set by
OPENAI_API_KEY	OpenAI secret key (starts with <code>sk-</code>). Used for GPT-4o and Whisper.	Owner (shared or per subscriber)

Push Notifications — Firebase

Variable	Description	Set by
FIREBASE_SERVICE_ACCOUNT	Full Firebase service account JSON as a single-line minified string. Includes <code>project_id</code> , <code>private_key</code> , <code>client_email</code> .	Owner (shared)

NZ Address Autocomplete — Addy.co.nz

Variable	Description	Set by
ADDY_API_KEY	Addy Solutions API key for NZ address autocomplete	Owner (shared or per subscriber)

Marketing — Meta (Facebook/Instagram)

Variable	Description	Set by
FACEBOOK_PAGE_ACCESS_TOKEN	Long-lived Page Access Token for subscriber's Facebook Page	Subscriber → Owner sets in env
FACEBOOK_PAGE_ID	Numeric ID of subscriber's Facebook Page	Subscriber → Owner sets in env
FACEBOOK_AD_ACCOUNT_ID	Numeric Ad Account ID from Facebook Business Manager	Subscriber → Owner sets in env

Authentication

Variable	Description	Set by
HERO_PASSWORD	Owner/admin password for the platform. Set this to a strong unique password per subscriber.	Owner (per subscriber)
SESSION_SECRET	Express session secret. Generate a random 32+ character string per deployment.	Owner (per subscriber)

7. Estimated Monthly Cost Summary

All costs are approximate and in NZD. Actual costs vary based on subscriber usage. These are **your costs** — what you pay to run the platform for one active subscriber per month.

Integration	Tier Required	Fixed Cost/mo (NZD)	Variable Cost	Notes
Neon PostgreSQL	All	\$0 – \$22	—	Free tier covers most small subscribers. Upgrade for more storage.
Resend Email	All	\$0 – \$22	—	Free for 3,000 emails/mo. Most subscribers stay on free tier.
Gmail	All	\$0	—	Free with standard Gmail account.
OpenAI	All	\$8 – \$25	Per token	Depends on AI chat and dispatch usage. Set spending limit per subscriber.
SMS Everyone NZ	All	\$0	\$0.12/SMS	Pre-purchased credits. 500 SMS = ~\$60. Estimate 100–300 SMS/month per subscriber.
Addy.co.nz	All	\$0	~\$0.015/lookup	~\$3–10/month per active subscriber.
Firebase FCM	All	\$0	—	Spark free plan: 1M notifications/month. Practically free.
Estimated total (Starter subscriber)		~\$50–120 NZD/mo		Excluding subscriber's own Xero/Mailchimp/Meta costs
Estimated total (Pro subscriber)		~\$80–150 NZD/mo		With Twilio, Xero, Calendar, Mailchimp

Integration	Tier Required	Fixed Cost/mo (NZD)	Variable Cost	Notes
Replit hosting	All	\$30 - \$60	—	Per Replit deployment. Share across multiple subscribers if on higher plans.
Twilio Voice	Pro+	\$4	~\$0.02/min	NZ number ~\$2.50 USD/mo. Call recording included. ~50 mins/month = ~\$1 USD extra.
Xero API	Pro+	\$0	—	API access is free. Subscriber pays for their own Xero plan (\$50-100/month directly).
Google Calendar	Pro+	\$0	—	Free.
Mailchimp	Pro+	\$0 - \$40	—	Free for <500 contacts. Subscriber's own account.
Meta API	Enterprise	\$0	—	API access is free. Subscriber pays their own ad budget to Meta directly.
Apple Developer	All (iOS)	\$1.75	—	\$149 USD/year amortised across all subscribers. One account covers all.
Google Cloud Storage	All	\$0 - \$10	Per GB stored	Included in Replit plan for small usage.
Estimated total (Starter subscriber)		~\$50-120 NZD/mo		Excluding subscriber's own Xero/Mailchimp/Meta costs
Estimated total (Pro subscriber)		~\$80-150 NZD/mo		With Twilio, Xero, Calendar, Mailchimp

□ **Margin example:** If you charge a Pro subscriber \$299 NZD/month and your costs are \$150 NZD/month, your gross margin is ~\$149 NZD/month per subscriber. At 10 subscribers: ~\$1,490 NZD/month recurring revenue above costs.

8. Ongoing Maintenance Notes

Token & Credential Renewal Schedule

Integration	Expiry / Renewal	Action required
Xero	Token expires every 30 min — auto-refreshed	No action. If scopes change in Xero Developer Portal, manually hit "Refresh Token" in platform Settings → Xero.
Twilio API Key	Never expires (unless manually rotated)	Rotate annually as security best practice. Update <code>TWILIO_API_KEY</code> and <code>TWILIO_API_KEY_SECRET</code> .
Gmail App Password	Never expires (unless 2FA is disabled or manually revoked)	If subscriber changes their Google password or disables 2FA, a new App Password must be generated.
Facebook Page Token	Expires in 60 days (short-lived) or never (long-lived System User token)	Set calendar reminder at 50 days to regenerate if using standard long-lived token. Switch to System User token for never-expiring tokens.
OpenAI API Key	Never expires	Monitor spending via OpenAI dashboard. Rotate if compromised.
Firebase Service Account	Never expires (key file)	Keep secure. Regenerate if compromised via Firebase Console → Service Accounts.
APNs Key	Never expires unless revoked	Keep the .p8 key file secure. Revoked keys break all iOS push notifications instantly. Do not delete from Apple Developer portal.
Resend API Key	Never expires	Rotate annually. Update <code>RESEND_API_KEY</code> in environment.
Google Calendar OAuth	Access token: 1 hour (auto-refreshed) Refresh token: never expires unless revoked	No action normally. If subscriber revokes access in Google Account settings, they must re-authorise via the platform.
Apple Developer Program	Annual renewal (\$149 USD)	Set calendar reminder 30 days before annual renewal date. Lapsed membership = App

Integration	Expiry / Renewal	Action required
		Store/TestFlight access revoked for all subscribers.

What to do when Xero token fails

1. Go to platform Settings → Xero
2. Click "Disconnect" → then "Connect to Xero" again
3. If scopes have changed: click "Refresh Token" and check the scopes shown
4. If credentials have been regenerated: update `XERO_CLIENT_ID` and `XERO_CLIENT_SECRET` in the subscriber's environment and restart the server

Platform update process

1. Test the update in your own development environment first
2. If schema changes are included, prepare migration instructions
3. Deploy to each subscriber's environment (Replit: pull latest → restart)
4. Run `npm run db:push` if there are schema changes
5. Notify subscribers of any changes that affect their workflow

SMS credit monitoring

SMS Everyone credits are pre-purchased and will run out without warning. To avoid service disruption:

- Check credit balance monthly via the SMS Everyone dashboard or API (`/api/credits`)
- Set a low-balance email alert in the SMS Everyone account settings
- Include SMS credits in your subscriber monthly fee (pre-purchase on their behalf) or bill them separately when credits run low

iOS app updates & TestFlight

- Build a new iOS binary when significant updates are ready: `npm run build` → `npx cap sync ios` → Archive in Xcode → Upload to App Store Connect
- TestFlight builds expire after 90 days — upload a new build before expiry
- App Store review required for public releases (1–3 days). TestFlight builds available within hours.
- For emergency patches, TestFlight is the fastest path — invite testers and distribute without App Store review

Address territory configuration

The address autocomplete is currently configured to search **Gisborne** territory only. When onboarding a subscriber in a different region:

- Edit `server/services/addressService.ts` line 27 — change `territory=Gisborne` to the subscriber's region or remove the parameter for nationwide search
- Valid NZ territories include: Auckland, Wellington, Canterbury, Waikato, Bay of Plenty, Manawatu-Whanganui, Otago, Gisborne, Nelson, Hawke's Bay, Taranaki, Southland, Marlborough, West Coast, Tasman

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